Carnegie Mellon University
Computer Science Academy

Description

Describe your program. What do you envision your app will do?

Note: At this step, do not be concerned about how to do it, just focus on what you envision.

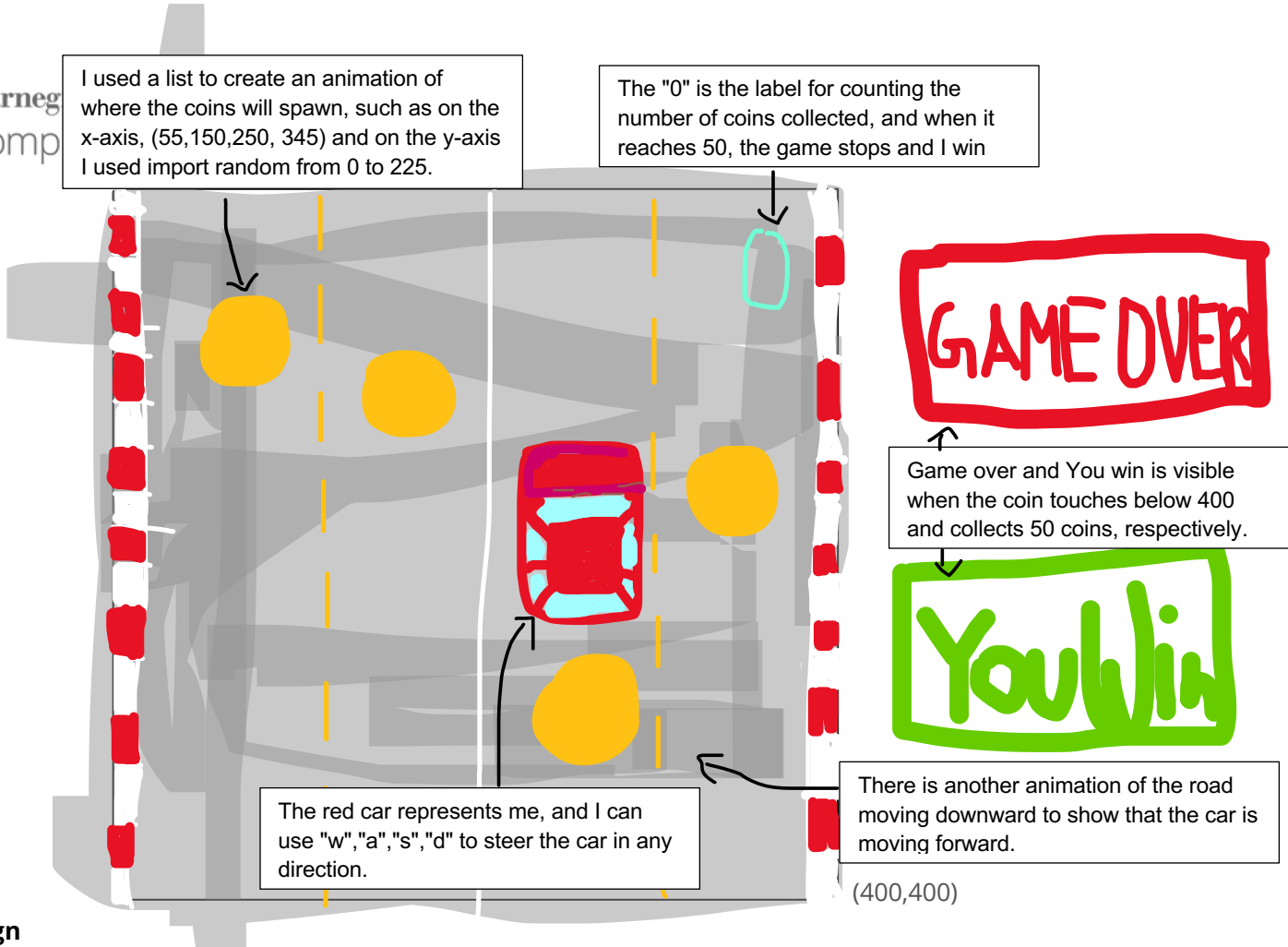
In this program, I've decided to create a racing game, but I realized it's going to be quite difficult for me to put all the stuff in a limited 400x400 canvas. So, I'll make a road with four lanes and sidewalks, and there will be lines drawn to separate these four lanes. Also, I want to make the road move downward, so I added it in onStep, but I found out the road is going to run out, so I increased the length of the road to ensure that it can last forever, and I used 'if statements" to help the roads go back to the top when they are below 400. Next, I'll create the car, which is drawn from the top-view and has many details of the windows. I want to add some obstacles to the course for the car to avoid, and if the car hits the obstacles the game will stop (I'll use app.stop()). Although, the obstacles weren't really interesting, so I actually decided to switch them to coins that the car can collect when hitting the coins. I also created a label on the top-right corner to record the number of coins the car has collected. I added onKeyHold commands to steer the vehicle. To make the game more interesting, I added "GAME OVER!" and "You Win!", respectively, when the coin is more than 400 on the y axis I lose, and when I collect 50 coins I'll win. Lastly, I limited the speed of the car to make the game more challenging.

Canvas Design

Sketch your canvas. Label any animation or user interaction.

- **Colors:** What colors will be dominant in your app? Are there specific shades that are important to the app's functionality or aesthetic?
- **Animations/Movements:** Will there be any animated elements? If yes, describe how you envision them. For example, will a shape move across the screen?

(0,0)



Code Design

Plan your code. List here the coding concepts and properties you plan to use to accomplish your goals, and why you plan to use them. These should incorporate at least two new concepts from this unit. Briefly describe what purpose each element will serve in your project. Aim for at least four to six properties.

Concept	Purpose
app.stop()	to end the program when the coin reaches below 400 or when there are 50 coins collected.
if statement	to check if an objects is equal to what I want it to do and then do something
Group	the car needs a lot of lines of code, so I will use Group to combine them altogether for easier use later
Import random	in order to use "random.randint" I will need to import random at the beginning
app.count	for the coins to spawn one at a time and give me an opportunity to adjust it's spawning speed

onKeyHold	I want to be able to move the car around, so I'll need to define some keys
onStep()	when the program runs, I would like to be able to see the road already moving downward and coins spawning
for loop	animation for the coin in the program to do what I want it to do
list	

Creative Task Reflection

After completing your creative task use this form to reflect on your work.

Difficulties and Opportunities

For this section, go in-depth into either a difficulty you faced or an opportunity you saw to improve or add to your code.

If you Faced a Difficulty:

- Describe the Problem: What specific issue did you encounter?
- Attempted Solutions: What did you try in order to solve the problem?
- Final Solution: What was the final solution that fixed the issue?

If you Found an Opportunity:

- Reason for the Addition/Change: What made you consider this opportunity? Was it a challenge or an aha-moment?
- Impact on Design: How did seizing this opportunity impact other aspects of your design?
- Design Trade-offs: What aspects of your original design did you have to adjust or remove to incorporate this new feature?

Something that was challenging during the creation of the program is fixing the speed of the obstacles spawning. I tried many different codes, such as I added "onsteppersecond" to try to modify the speed, but it doesn't work. Mr. Stephens showed me a way of how to personally modify the speed of the spawning coins and how it drops one by one.
If I can have another opportunity to enhance this program, I would like to create a two-person game. With another car that can be driven on the right side of the canvas, and two people must compete to see who can collect the most coins in the limited amount of time. Although, I did make some useful changes to my program, for example, by adding "GAME OVER" and "You Win!". This change significantly increased the

interest of my program, and it's more fun to play with a fear of losing and the motivation to win.

Finished Project Code Outline

List **only one instance** of each important coding concept used in your project, along with the corresponding line number, and describe how it contributes to the overall function of your program. **Highlight the aspect you are most proud of.** For each concept, also include a brief definition to demonstrate your understanding. By focusing on one instance of each concept, aim to provide more depth and clarity in your responses.

Concept	Line number(s)	Definition	Purpose
import random	1, 81	It enables the program to perform actions that seem unpredictable	in order to use "random.randint" I will need to import random at the beginning
Rect	2, 43, 48,	it draws a rectangle	to draw the background gradient and the 'GAME OVER' and 'You Win!' board.
Line	4-6,	draws a straight line from one point to another	to draw the lines on the road to separate the lanes
Group	7-39,42-51	to combine multiple lines into one big group	to combines the codes that draws the car, the win & lose, the sidewalk, and the road into one group, respectively.
app.count	40, 79, 99	custom property that counts something	for the coins to spawn one at a time and give me an opportunity to adjust it's spawning speed
def onKeyHold	52-68	it defines what happens when the key is held	it moves the car in any directions
if statement	53-68,74-98	executes a block of code if its condition is True	to check if an objects is equal to what I want it to do and then do something
def onStep()	72-103	enables the codes under this definition to run after the program begins	when the program runs, the road will be moving downward and coins spawning
for loop	58-98	a control flow statement that is used to repeatedly execute a group of statements as long as the condition is satisfied	animation for the coin in the program to do what I want it to do

list	80	a built-in data type for storing ordered collections of items	I used "choice" to tell the program where to spawn the coin on the x-axis

Future goals

Reflect on your entire design and coding process, from conceptualization to implementation. In this section, consider not only what you would change in the final product but also how you would approach the entire project differently.

- **Code Changes:** What specific changes would you make to your code if you had more time or if you were to start from scratch? Would you add new features or refine existing ones?
- **Process Adjustments:** Did you stick to the original plan, or were there deviations? Were these changes beneficial or did they lead to complications? How could your planning or approach be improved?
- **Resource Utilization:** Did you make the best use of available resources, like class notes, tutorials, or peer help? What additional resources could have been beneficial?
- **Time Management:** Reflect on how effectively you managed your time. What would you do differently to ensure a smoother process?

By considering these aspects, you will not only think about 'what' you would change but also 'how' and 'why,' which can be incredibly enlightening for future projects.

If I had more time, I would like to enhance the program to make it a two-person game instead of a one-person game because it would be more fun to let friends play too and compete. When I almost finished this program, I thought of another very cool program I would have created from scratch, which is creating a stick man standing at the bottom of the canvas and walking only left and right to catch as many candies as he can in a limited time. I didn't stick to my original play, which is to create a racing game and avoiding obstacles because it's way too hard to create. The changes have significantly impacted the result of the program because I changed the obstacles from rocks to coins. This change benefited a lot because I got to create a new game that is the car needs to collect as many coins as possible to win. I asked the teacher for help if I really need the help and asked my classmates to test my program to see if some features are suitable for playing. Next time, I would like to plan faster and brainstorm on more ideas of what the program will look like because I'm running out of time as it comes to the last second before the semester ends.

Scroll down for submission checklist.

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Pre-Submission Checklist for Creative Tasks

Program Design

- [ ] Described what your program will do clearly.
- [ ] Incorporated at least two concepts from the current unit.
- [ ] Added annotations to explain colors, movements, and user interactions.

Canvas Design

- [ ] Sketched your canvas layout.
- [ ] Labeled any animation or user interaction on the canvas sketch.
- [ ] Detailed dominant colors and their importance.
- [ ] Described any animations or movements.

Code Design

- [ ] Listed at least four to six coding concepts or properties you plan to use.
- [ ] Explained why each concept or property is essential for your program.

Program Development and Coding

- [ ] Finished coding the program.
- [ ] Debugged and tested the program.
- [ ] Checked if the program meets the design criteria.

Difficulties and Opportunities

- [ ] Described a difficulty faced or an opportunity realized in detail.
- [ ] Explained the steps taken to resolve the difficulty or seize the opportunity.

Finished Project Code Outline

- [ ] Listed only one instance of each important coding concept used.
- [ ] Provided the line numbers for each coding concept.
- [ ] Included a brief definition for each coding concept.
- [ ] Highlighted the aspect you are most proud of.

Future Goals

- [ ] Discussed what specific changes you'd make if given more time.
- [ ] Reflected on your approach and planning for the project.
- [ ] Considered resource utilization and time management.

Final Check

- [ ] Revisited the rubric to make sure you meet all the criteria.
- [ ] Made sure your document is well-organized and free of grammatical errors.
- [ ] Ready to submit the project and this document.

By ticking off each item on this checklist, you ensure that you have met all the requirements and can confidently submit your work for evaluation.